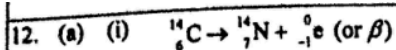


- | | | | | | |
|----|-----|------|---|----|----------|
| 6. | (a) | (i) | Accelerates at g before the elastic cord stretches / at the beginning. | 1A | |
| | | | Acceleration decreases as the cord stretches. | 1A | |
| | | | Decelerates until momentarily at rest | 1A | |
| | | | (after the tension in the cord is greater than mg). | | <u>3</u> |
| | | (ii) | Gravitational potential energy changed to kinetic energy and | 1A | |
| | | | (then) elastic potential energy in elastic cord. | 1A | <u>2</u> |
| | (b) | | Elastic cord lengthens the stopping time, | 1A | |
| | | | hence reduces the (net) force acting on the player. | 1A | <u>2</u> |
| | (c) | | Contact area is larger, | 1A | |
| | | | hence pressure is smaller during the fall and the structure is less likely to break / detach. | 1A | <u>2</u> |

10. (a)	Alpha particles emitted can be stopped by the (thin) metallic casing.	1A
	<u>Or</u> Shorter range / Lower penetrating power.	1A
		1
(b) (i)	$k = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{87.74 \times 3.16 \times 10^7}$ $= 2.5 \times 10^{-10} \text{ s}^{-1} \text{ or } 7.9 \times 10^{-3} \text{ year}^{-1}$	1M
	Activity $A = kN$	
	$= \frac{\ln 2}{87.74 \times 3.16 \times 10^7} (3.2 \times 10^{25})$ $= 8.000 \times 10^{15} \text{ (Bq)}$	1M 1A
		3
(ii)	Power = Energy per decay $\times$ Activity $= 5.5 \text{ MeV} \times 8.000 \times 10^{15} \text{ Bq}$ $= 5.5 \times 10^6 \times 1.60 \times 10^{-19} \times 8.000 \times 10^{15}$ $= 7040 \text{ W or } 7.040 \text{ (kW)}$	1M 1A
		2
(iii)	Power $\propto$ Activity Activity $\propto N$ $\therefore$ Percentage of power left $= \left(\frac{1}{2}\right)^{t/t_{1/2}} \times 100\%$ $= \left(\frac{1}{2}\right)^{36/87.74} \times 100\%$ $= 75.25\% \approx 75\%$	1M 1A
	<u>Or</u> Percentage / fraction of power left $= 3/4$	
	<i>Alternatively:</i> $N = N_0 e^{-kt}$ $\therefore$ Percentage of power left $= e^{-kt} \times 100\%$ $= e^{-(\ln 2 / 87.74) \times 36} \times 100\%$ $= e^{-0.2844} \times 100\%$ $= 75.25\% \approx 75\%$	1M 1A
		2

10. (a)	(i)	226 - 206 = 20 (multiple of 4 for α) $\therefore {}^{206}_{82}\text{Pb}$ is the end product	1M	Accept 97.8 % to 98.0 %	
			1A		
		2			
	(ii)	% undecayed Ra-226 left $\frac{N}{N_0} = e^{\frac{\ln 2}{1600} \times 50}$ $= 97.857\%$ $\approx 97.9\%$	1M		
			1A		
			Or % undecayed Ra-226 left $= \left(\frac{1}{2}\right)^{\frac{50}{1600}}$ $\approx 97.9\%$		1M
					1A
		2			
	(b)	(i)	$\therefore$ random process		1A
					1
		(ii)	Some of the daughter products of Ra-226 are also radioactive and may emit β particles.		1A
					1
		(iii)	Reason: weaker ionizing power of β and γ radiations - raise the source to a distance greater than the range of α (a few cm) will cease to have sparks. - insert a paper between the source and the gauze, sparks will cease to produce.		1A
					Any ONE
					1A
					2

9.	(a)	(i)	β decay / beta decay OR ${}_{19}^{40}\text{K} \rightarrow {}_{20}^{40}\text{Ca} + {}_{-1}^0\beta$	1A	
				1	
		(ii)	Justified: the penetrating power of β radiation enables it to penetrate the body's organ / skin Or Not justified: the activity is low and is comparable to background radiation / β radiation is largely shielded by the human body	1A	
				1	
(b)	(i)	$\frac{0.45 \times 0.012\%}{40.0} = 1.35 \times 10^{-6}$ (mole)	1A	1	
	(ii)	$k = \frac{\ln 2}{1.25 \times 10^9 (3.16 \times 10^7)} = 1.754803 \times 10^{-17} \text{ (s}^{-1}\text{)}$	1M/1A		i. e. $5.545177 \times 10^{-10} \text{ (year}^{-1}\text{)}$
		Activity = kN $= 1.754803 \times 10^{-17} \times (1.35 \times 10^{-6} \times 6.02 \times 10^{23})$ $= 14.261284 \text{ (Bq)} \approx 14.3 \text{ (Bq)}$	1A	2	Accept 14.2 to 14.3 (Bq)

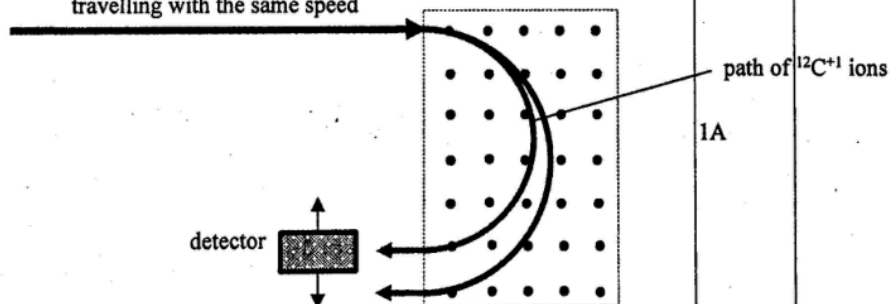


(ii) $k = \frac{\ln 2}{t_{\frac{1}{2}}}$
 $= \frac{\ln 2}{5370}$
 $= 1.2097 \times 10^{-4} \text{ (yr}^{-1}\text{)} \approx 1.21 \times 10^{-4} \text{ (yr}^{-1}\text{)}$

(iii) $C = C_0 e^{-kt}$
 $6.0 = 13.6 e^{-(1.21 \times 10^{-4})t}$
 $t = 6764.7 \approx 6800 \text{ (years)}$

(b) (i) An isotope is a variation of an element that possesses the same atomic number but a different mass number.

(ii) ${}^{12}\text{C}^{+1}$ and ${}^{14}\text{C}^{+1}$ ions
travelling with the same speed



(iii) As the magnetic force induced on a carbon ion is always perpendicular to its velocity / direction of motion, no work is done on the ions.

1A

1

1M

1A

2

1M

1A

e.c.f. from (a)(ii)

Accept: 6760 ~ 6820 (years)

Or

$$(13.6) \left(\frac{1}{2} \right)^n = 6.0$$

$$n \log \left(\frac{1}{2} \right) = \log \left(\frac{6.0}{13.6} \right)$$

$$n = 1.181$$

$$t = n \times 5730 = 6764.7 \text{ (years)}$$

$$\approx 6800 \text{ (years)}$$

2

1A

Or

same number of protons but
different number of neutrons

1

1A

1

1A

1A

2